

PINAK GUPTA

☎ +91 8264558604 ✉ pinakgupta99@gmail.com 🔗 LinkedIn 🐙 GitHub 🌐 Portfolio

EDUCATION

Dr. B. R. Ambedkar National Institute of Technology , Jalandhar Nov 2022 - Present
Bachelor of Technology in Computer Science and Engineering - **CGPA: 9.05/10** Jalandhar, India

Dayanand Model Sr. Sec. School , Jalandhar May 2020 - June 2022
Senior Secondary Education (CBSE) - **Percentage: 92.2%** Jalandhar, India

EXPERIENCE

Research Intern - Robotics and Artificial Intelligence June 2025 - Present
Indian Institute of Technology, Mandi Mandi, India

- Developed autonomous ball-hitting system using **Kinova Gen3** robotic arm with racket for tennis ball hitting.
- Fine-tuned **YOLO11** model for real-time yellow tennis ball detection achieving **99.5%** mAP@0.5 accuracy.
- Designed **3D** ball trajectory prediction system using **Kalman Filters** and physics-based modeling with **Depth Anything v2 Vision Transformer** model (**97.5M** parameters).
- Trained end-to-end **reinforcement learning agent** using **Proximal Policy Optimization (PPO)** in **MuJoCo** simulation environment enabling successful ball interception and striking behavior.

PROJECTS

OneVote | *Express.js, Node.js, React.js, MongoDB, TypeScript, Redux Toolkit, Resend* 🐙 GitHub

- Built secure **MERN**-based voting platform with multi-election support and responsive **UX** for elections.
- Implemented multi-factor authentication with email OTP via **Resend API**, **JWT** authentication.
- Developed admin dashboard with candidate CRUD operations, **voter whitelisting**, and result visibility controls

Welth - AI Finance Platform | *Next.js, Supabase, Tailwind CSS, Clerk, Gemini AI* 🐙 GitHub

- Architected finance platform with multi-account tracking, **AI-powered** categorization, visualization dashboards.
- Implemented **Clerk authentication** and **Gemini AI** for automated expense scanning.
- Engineered financial workflows using **Inngest** for recurring transactions and AI-driven analysis via **Resend**.

QuickBite | *MongoDB, Express.js, Node.js, React, Stripe, Tailwind CSS* 🐙 GitHub

- Built food delivery platform using **MERN** stack with location-based search by city and cuisine for management.
- Integrated **Auth0 authentication** and **Stripe payment processing** for transactions.
- Developed real-time order tracking system allowing restaurants to update status from preparation to delivery.

ACHIEVEMENTS

- Earned **Guardian** badge on **LeetCode** 🏆 with peak rating of **2143**, solving **1000+** algorithmic problems.
- Achieved rank of **685** among **30K** participants in LeetCode Weekly Contest **457**, demonstrating problem-solving.
- Attained **4-Star** programmer status on **CodeChef** 🏆 with maximum rating of **1949**.
- Secured rank of **85** in CodeChef Starters **171** (Division 2), showcasing competitive programming capabilities.
- Achieved **Expert** status on **Codeforces** 🏆 with peak rating of **1667**.
- Ranked **1192** in Codeforces Round **1002** (Div.2) demonstrating algorithmic proficiency and speed under pressure

SKILLS

Languages: C/C++, Python, Java, HTML, CSS, JavaScript, SQL

Technologies: MongoDB, Express.js, React.js, Node.js, Tailwind CSS

Courses: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOPS), Database Management Systems (DBMS), Machine Learning, Computer Networks (TCP/IP), Operating System

EXTRA-CURRICULAR ACTIVITIES

Core Team Member, Competitive Programming Oct 2023 - Present
Google Developer Groups

- Enhanced expertise in **Data Structures and Algorithms (DSA)** through competitive programming, significantly honing problem-solving skills and contributing to community growth.